HASSAN ASIF

(+92) 302-043-3490 | hassanop1122@gmail.com
Lahore, Pakistan

PROFILE

Embedded Systems & IoT Engineer specializing in industrial monitoring and automation solutions. Experienced in designing and deploying ESP-based systems integrating Modbus sensors, RF communication, and GSM networks for real-time data acquisition and control. Proven ability to build end-to-end systems from hardware design to cloud integration.

CORE SKILLS

Embedded Systems (ESP32, Arduino) | IoT Architecture | Modbus Integration | GSM & RF Communication | Circuit Design & Debugging | Industrial Automation | Python, C,JAVA, MATLAB | Proteus, LTspice

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PROFESSIONAL REGISTRATION

Registered Engineer — Pakistan Engineering Council (PEC)

EXPERIENCE

Tracking World Pvt Ltd — Embedded Systems Developer (R&D)

Successfully deployed IoT-based industrial monitoring systems in live field environments

Currently monitoring real-time operational data remotely via cloud with continuous operation for 2+ months

Integrated Modbus sensors, RF modules, and GSM communication for reliable industrial data acquisition

Developed complete solution including hardware design, firmware, and cloud dashboard integration

Reduced manual monitoring requirements through automation and remote visibility

Nishat Mills Limited — Electrical & Electronics Intern

Worked on troubleshooting and maintenance of industrial electrical systems.
Gained exposure to automation systems and control panel operations in production environment.

Looptex — Electrical Intern

Performed wiring, load calculations, and equipment monitoring. Supported preventive maintenance and system diagnostics.

EDUCATION

BS Electrical Engineering (Electronics) — COMSATS University Lahore (2021–2025)

FSC Pre Engineering — Punjab Group of Collages (2019-2021)

Matric (Science) — Bourn Hall High School (2017-2019)

KEY PROJECTS

Industrial Environmental Monitoring System

Designed IoT system measuring **Temperature, Humidity, CO_2, and AQI**

Operating continuously in field for **2+ months** with stable real-time data reporting

Enabled remote cloud monitoring and reduced manual inspection

Displaying on **P10 led panels**

Automatic Tank Gauge (ATG)

Developed wireless fuel monitoring system using Modbus sensors and RF communication. Designed sender/receiver architecture with cloud integration for remote tracking.

Designed **Dgus Dwin Lcd**

Generator Remote Monitoring System

Built **GSM-based generator monitoring solution** for status and fuel tracking

Running continuously for 2+ months with real-time monitoring and reporting

Integrated energy meters for automated data collection

Final Year Project: Empowering Customer Service Robot

Developed a general-purpose service robot with **advanced interaction features** .

Focused on smart interaction through hardware/software integration .

Ultrasonic Radar System

Created a radar system using ultrasonic sensors and servo motors for object detection and scanning

Smart Office Using IR Sensors

Designed a smart office automation prototype to control lights and appliances based on occupancy .

PID Controller

Implemented a **Proportional-Integral-Derivative** control system for speed and **stability**

Soft Skills:

Team Management, Leadership, Interpersonal Communication, Technical Documentation

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Languages

1. Urdu
2. English